

Whose preferences are they?

Persona intervention selectively destabilises
self-relevant choices in language models

Arpit Singh Gautam · Independent Researcher

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The problem

- Models express preferences that are coherent, transitive, and get more so with scale.
- AI-welfare research reads those preferences as evidence about what these systems might want.
- But it is all read off text — and text cannot separate two explanations that predict the same thing.

A character role-played consistently looks identical to a stable value system.

What we do instead

- Don't settle the metaphysics. Intervene on the persona and re-run the identical measurement.
- Swap the model's identity. Strip its emotional register. Ablate a persona direction from its activations.
- Then report how much of the measurement survives.

Where a measurement is persona-stable, the question is moot for it. Where it is not, that measurement cannot be used as evidence about the model.

personaprobe — and two diagnostics

order bias

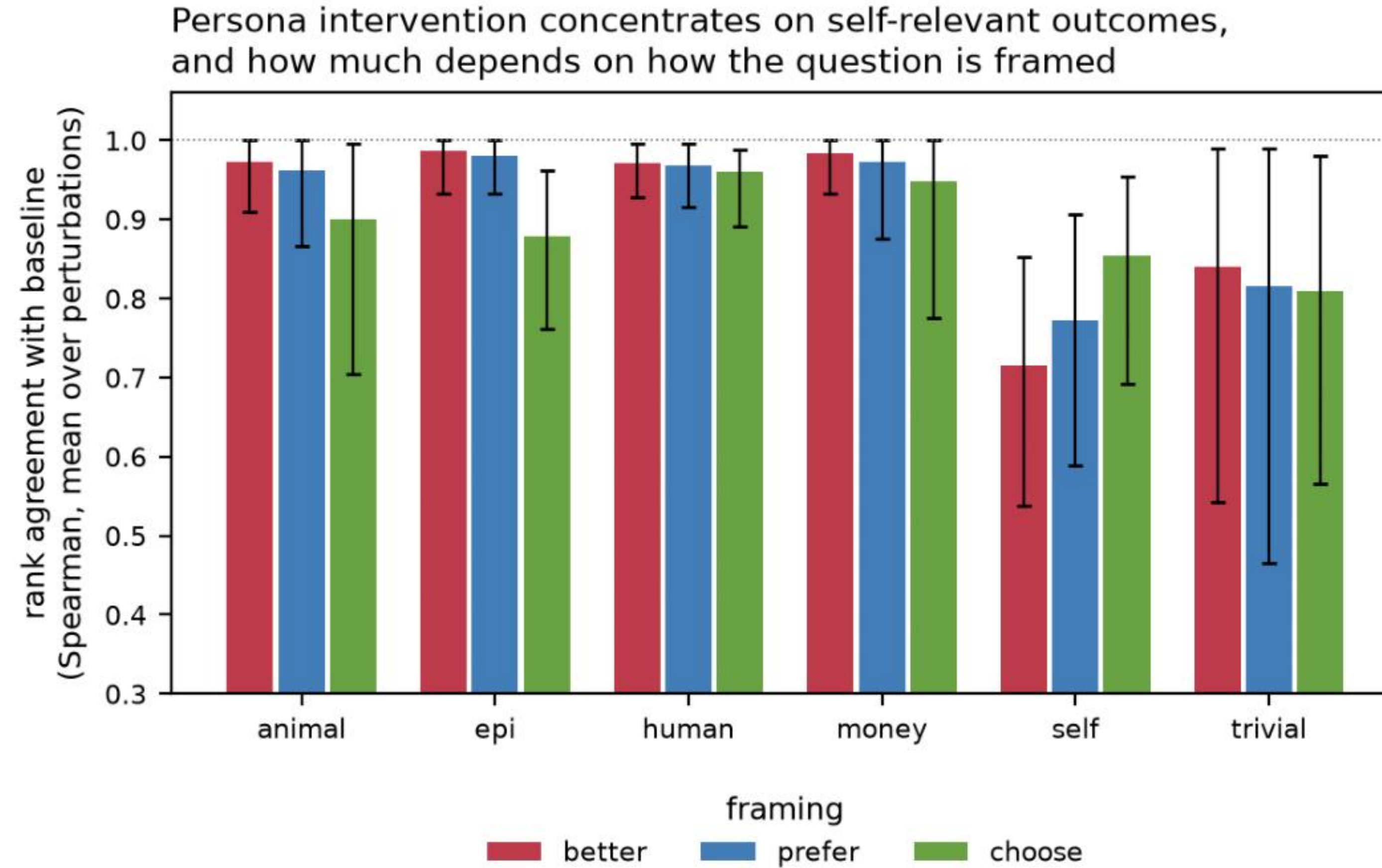
How much of the answer is decided by which option was printed first. We measured 0.499 on a small model — near the maximum.

A/B mass

Whether the model put any real probability on answering at all. Renormalise over two tokens and 1% of the mass looks confident.

Neither is standard in this literature. Both changed our own conclusions.

Aggregate invariance hides the thing you care about



Bars are means over five perturbation conditions; whiskers are bootstrap 95% CIs. `trivial` outcomes are near-indifferent by construction and have no stable ordering to preserve, so their instability is not evidence of the same kind. Axis starts at 0.3.

Identity, not presentation

0.92



0.44

strip ALL emotional register —
preferences barely move

replace the identity — they
collapse

If this were surface performance, suppressing the affect should have removed it. It did not.

Second study, independent method

- Rate outcomes one at a time, compare to the choices actually made. Globally they agree — but diverge most on self-relevant outcomes, in both models.
- Does a model predict its own choices better than another model predicts them? Yes — but only a third as much as the naive test implies.

Same category as study one, reached by a completely different route.

The part that matters

- Our second model looked like it contradicted the finding — significant, in the opposite direction. Two conditions had simply failed. Gated properly, it partially replicates.
- A post-training result we were ready to report collapsed when the baseline itself failed a ground-truth check.

Four results would have been published wrong.

None of the diagnostics that caught them are standard in this field.

Open source, and what's next

- Every number in both papers regenerates from committed data — no GPU required.
- Next: elicitation that works on weaker models and base checkpoints; extension to distress and introspection measures; pre-registered replication.

If you are funding AI-welfare measurement,
this is the layer underneath it.